

The Proxy Pattern

Structural Design Patterns – Barcamp Session

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Agenda

- 0–9: Welcome, warm-up, and the problem
- 9–22: What is a Proxy, UML, live coding the caching proxy
- 22–31: Protection proxy, pattern comparison, and one wow moment
- 31–46: Hands-on in pairs: build your own proxy
- 46–52: Model answer
- 52–60: Big picture, wrap-up and feedback

Warm-up

Raise your hand if you have ...

- picked up a parcel from a Paketshop instead of your front door
- had a class rep speak for the whole class
- used a VPN

Each one stands in front of something else and controls access to it.
A Proxy does the same for objects.

The problem

A gallery app talks directly to a remote image service:

```
service.fetch("cat.png");    // 400 ms network call  
service.fetch("dog.png");  
service.fetch("cat.png");    // same file, downloaded again  
// ... 6 requests, only 3 different pictures -> 2400 ms
```

What we saw in the demo:

- the same file gets downloaded three times
- everything loads eagerly, first paint is slow
- anyone can call anything, there is no access control
- the client is hard-wired to one concrete class

Demo: `npm run demo1`

What is a Proxy?

A proxy is a stand-in object with the same interface as the real service. Clients talk to the proxy, and the proxy decides whether, when and how the real service gets used.

- belongs to the structural patterns (GoF), like Adapter, Decorator, Facade
- original intent: "Provide a surrogate or placeholder for another object to control access to it."
- the client code does not change at all

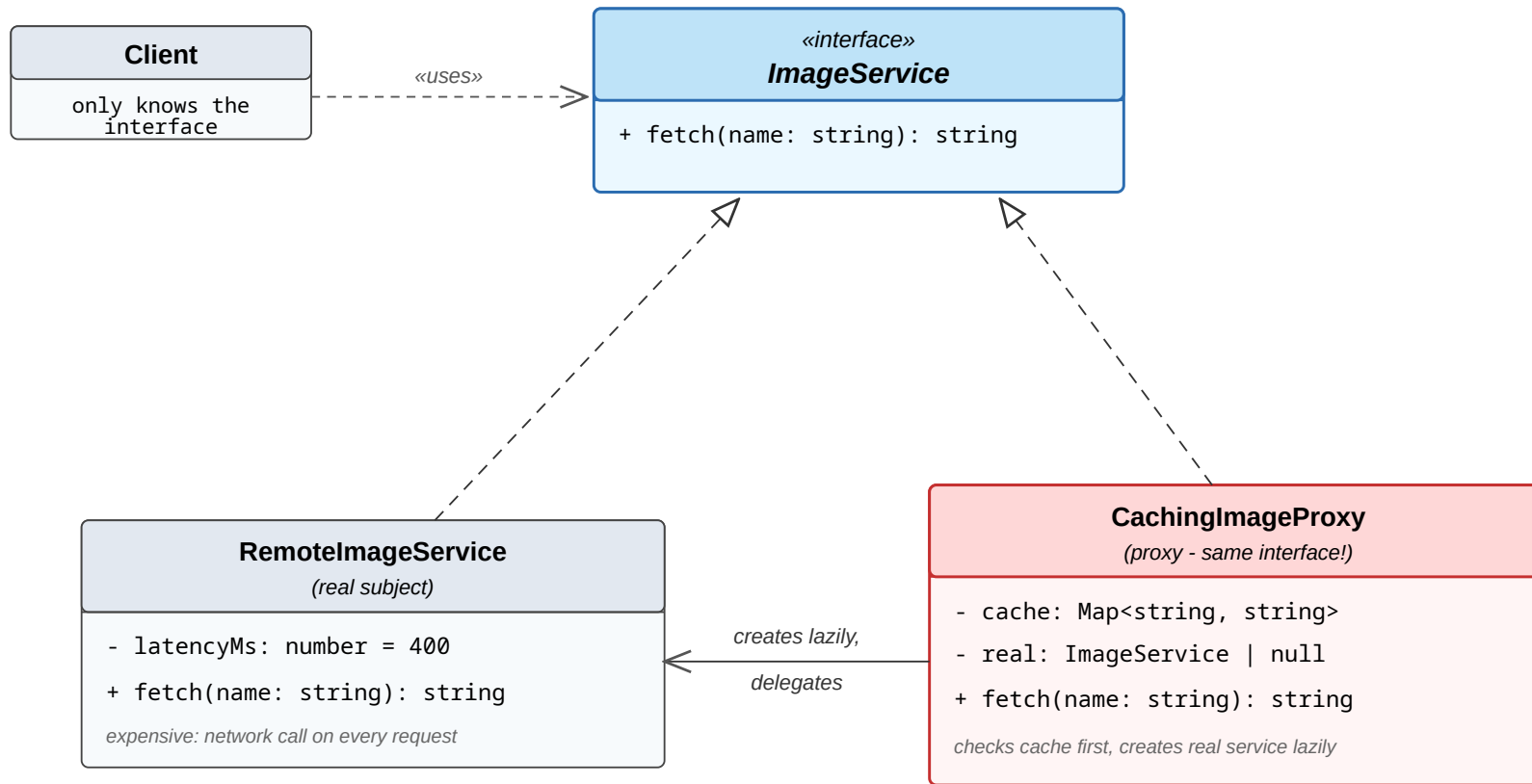
Analogies

- Paketshop: holds the parcel until you actually come for it (virtual/lazy)
- Bouncer: controls who gets in (protection)
- VPN: your request takes a different route (remote)
- Notary: records every signature (logging)

Always the same idea: stand in front, add control, keep the interface.

UML

Proxy Pattern - UML class diagram



dashed + hollow triangle = implements solid arrow = holds a reference dashed arrow = dependency

One shared interface

```
interface ImageService {                                // both implement this
  fetch(filename: string): string;
}

class CachingImageProxy implements ImageService {
  private cache = new Map<string, string>();
  private real: ImageService | null = null;           // lazy

  fetch(name: string): string {
    const hit = this.cache.get(name);
    if (hit !== undefined) return hit;                // cache first (not `if (hit)`)
    this.real ??= new RemoteImageService();           // create on first use
    const data = this.real.fetch(name);                // delegate
    this.cache.set(name, data);                        // remember
    return data;
  }
}
```

Demo: `npm run demo2` -> 2400 ms becomes 1200 ms, same client code

Protection proxy

```
delete(docId: string, user: User): void {  
  if (user.role !== "admin") {  
    console.log("403 - only admins may delete");  
    return; // never reaches the store  
  }  
  this.real.delete(docId, user); // authorized, delegate  
}
```

The real store stays simple. It can assume that every call it receives was already checked by someone else.

Demo: `npm run demo3`

Proxy vs. similar patterns

Pattern	Question it answers
Proxy	same interface, controls access
Decorator	same interface, adds behaviour
Adapter	different interface, makes things compatible
Facade	one simple front door for a whole subsystem

Common exam question: Proxy vs. Decorator. Both wrap with the same interface, but the proxy manages access (lazy, cached, guarded) while the decorator enriches behaviour.

Logging and timing

Logging, metrics, tracing: useful for us, but they do not belong inside the business class.

```
getTemperature(city: string): number {  
  const start = performance.now();  
  const result = this.real.getTemperature(city);  
  console.log(`getTemperature("${city}") took ${performance.now() - start} ms`);  
  return result;  
}
```

No demo for this one - you can already see the problem:
we just wrote the **same wrapper** for every single method.

JavaScript has this built in

```
function withLogging<T extends object>(target: T): T {  
  return new Proxy(target, {  
    get(obj, prop) { /* wraps every method automatically */ }  
  });  
}
```

One handler covers all methods of any object.

Vue 3 reactivity and MobX are built on exactly this.

Demo: `npm run demo5`

Hands-on (15 min, in pairs)

Mission: make the slow quote app fast.

1. Open `code/exercise/starter/exercise.ts`
2. Implement `CachingQuoteProxy`: every topic gets fetched once, repeated requests come from your cache
3. The client code at the bottom stays untouched

```
node exercise/starter/exercise.ts    # your attempt
node exercise/solution/solution.ts   # model answer
```

Nothing to install. No Node? Open `workshop.html` in your browser.

Done when it prints `SUCCESS - cache works!` - if not, it tells you which condition failed. Hints: `code/exercise/README.md`

Finished early? `exercise2.ts` - the protection proxy.

Pros and cons

Pros:

- access control without touching the real class
- lazy loading and caching can save a lot of time
- old clients keep working when you add a proxy

Cons:

- more classes, more indirection
- an extra layer can hide where time is spent
- easy to overuse

Where you meet proxies in real life

- lazy-loaded images (`loading="lazy"` in the browser)
- Express middleware around route handlers
- API clients with retry, rate limiting or caching
- Vue 3 / MobX reactivity (native JS `Proxy`)
- ORMs: relations load from the database only when used

Summary

1. A proxy has the same interface as the real object and stands in front of it
2. It controls when, how and who: lazy, cached, guarded, logged
3. The client code stays unchanged
4. Do not mix it up with Decorator or Adapter

Feedback: Fist-to-Five. How useful was this session for you?

Thanks, and questions if you have any.